Software Engineering Report

# PizzaHút Online Ordering System

## Sections 2 and 3 - Planning and Analysis

Focus: Project Feasibility, Initial Plan, and Use Case Analysis for the Web MVP

March 22, 2026

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# 1 Planning

## 1.1 Initial System Request

### 1.1.1 Business Need

PizzaHUST addresses a common problem in small food businesses: ordering and store operations are still handled manually through phone calls, in-person requests, and ad hoc coordination between the front desk, kitchen, and delivery process. This causes delays, inconsistent order tracking, limited visibility into promotions and menu performance, and unnecessary effort for the store owner. The proposed system digitizes the full web ordering flow while also supporting menu management, kitchen coordination, delivery handoff, and basic operational reporting.

### 1.1.2 Business Requirements

Key functional and non-functional requirements of the proposed system include:
Functional Requirements:

- Allow guests and registered customers to browse menus and view item details online.
- Support pizza customization by size, crust type, and optional toppings.
- Support side dishes, combo campaigns, and category-based menu browsing.
- Provide cart management and checkout with cash on delivery only.
- Allow guests to track orders using an order code.
- Allow registered customers to log in, manage profile data, view order history, and use loyalty points.
- Provide an admin area for managing pizzas, toppings, side dishes, categories, combo campaigns, customers, orders, and reports.
- Provide a kitchen order queue that helps kitchen staff process urgent orders first and update preparation progress.
- Integrate with a third-party delivery service for delivery booking and delivery status synchronization.
- Provide basic sales and order reports for the store owner.

Non-functional Requirements:

- The web interface must be responsive and usable on both desktop and mobile browsers.
- The system architecture must remain simple enough for a student team to deliver within the available academic timeline.
- The database must support customizable pizza orders, combo structures, loyalty points, and order tracking without forcing major redesign later.
- The codebase and interfaces must be clear enough to allow backend, frontend, and database work to proceed in parallel after the design phase.

### 1.1.3 Business Value

PizzaHUST provides clear business and academic value:

- **Improved Customer Convenience:** Customers can browse, customize, order, and track pizza deliveries online instead of relying on manual ordering channels.
- **Operational Efficiency:** The store owner can manage products, combo campaigns, and order flow through a centralized system.
- **Better Kitchen Coordination:** The kitchen order queue reduces ambiguity in deciding which order should be processed next.
- **Visibility and Reporting:** Basic reporting helps the shop monitor revenue, order counts, and item performance.
- **Strong Course Fit:** The system provides a realistic software engineering case with user-facing flows, back-office operations, data modeling, and external API integration.

### 1.1.4 Special Issues or Constraints

Several constraints shape the first version of the project:

- **Platform Constraint:** Version 1 is limited to a web application.
- **Payment Constraint:** The system supports cash on delivery only; no online payment gateway is included in the MVP.
- **Delivery Constraint:** Delivery is handled through a third-party delivery service rather than an internal shipper portal.
- **Scope Constraint:** AI-based menu recommendation is excluded from the MVP and treated as a future enhancement.
- **Time Constraint:** The team must prioritize design and integration risk early because the remaining implementation roadmap is limited to 10 weeks after the planning and general use case stage.

## 1.2 Feasibility Analysis

### 1.2.1 Technical Feasibility

**Current Technology Availability:** The system can be implemented using a conventional web stack that is already well supported by documentation and tooling. Next.js and Tailwind CSS provide a practical frontend solution for responsive screens, while FastAPI can handle API endpoints and business logic efficiently. MySQL is suitable for order, catalog, combo, loyalty, and reporting data. Docker-based deployment is sufficient for both development and demo environments.

**Developer Expertise:**

- **Familiarity with the application:** The team already understands the general online ordering scenario and has refined the MVP use cases, especially around guest

checkout, customer account features, admin management, and kitchen order queue handling.

- **Familiarity with the technology:** The team has some prior experience with web development technologies. Additional effort is still required to align the frontend, backend, and data contracts, but the technical learning curve is manageable for the chosen stack.

**Infrastructure Needs:**

- **Servers:** A lightweight VPS or cloud environment is sufficient for demo deployment because the expected traffic is small in the academic phase.
- **External Integration:** The main external dependency is the third-party delivery API. A mock integration path should be prepared to reduce schedule risk if the real API is unstable or unavailable.
- **Data Handling:** Standard authentication, role separation, and secure storage of user and order data are required, but no unusually complex infrastructure is needed for the MVP.

**Project Size:** The system is medium in academic scope. It includes a public ordering flow, customer account features, admin management, kitchen operations, delivery integration, and reporting. This is feasible for a 5-person student team if the design is stabilized early and the implementation stays within MVP boundaries.

**Compatibility:** Because the first release is web-only, compatibility focuses on modern desktop and mobile browsers. This is achievable with the selected frontend stack and avoids the additional overhead of native mobile development.

### 1.2.2 Operational Feasibility

**Ease of Use:** The customer-facing side is straightforward: users browse the menu, configure pizzas, manage the cart, and complete checkout. Guests can still place orders and track them using an order code, while registered customers gain order history and loyalty features. On the internal side, the admin and kitchen views are separated by role, reducing confusion and helping staff focus on their daily tasks.

**Maintenance:** Maintenance is manageable because the architecture remains a modular monolith rather than a distributed system. Admin catalog changes, combo schedules, and reporting queries can be updated incrementally. The largest operational maintenance risk is the delivery integration, which is why a mock-first approach is recommended during development.

**Workforce Requirements:** The workload is feasible for the current team because responsibilities can be divided into architecture and integration, backend order logic, UI/UX and frontend flow, database design, and acceptance-flow review. The clarified role of Nguyen Xuan Chi Thanh as Product/UX Support further strengthens the design stream instead of leaving UI work to one member alone.

### 1.2.3 Economic Feasibility

**Development Cost:** For an academic project, the direct cost is mainly a notional estimate of implementation effort rather than a commercial budget. A reasonable estimate for the development value of this MVP is **75,000,000 to 90,000,000 VND**, reflecting UI/UX design, database design, implementation, testing, integration, and deployment preparation.

**Infrastructure Cost:**

- **Development and Demo Phase:** Approximately 0 to 1,000,000 VND per month.
- **Stable MVP Deployment:** Approximately 1,000,000 to 5,000,000 VND per month depending on hosting and usage.

**Expected Benefits:** The system reduces manual effort, improves customer convenience, centralizes menu and order management, and provides basic operational visibility. For the course, it also delivers strong analysis, design, implementation, and integration value within a realistic business scenario.

### 1.2.4 Schedule Feasibility

The project remains feasible if the team avoids scope creep and keeps the implementation aligned with the refined MVP. The riskiest areas are checkout logic, kitchen order queue prioritization, delivery integration, and data modeling for customization and loyalty points. To control these risks, the next two weeks are dedicated to complete UI/UX coverage, ERD design, architecture decisions, and flow specifications before heavy implementation begins.

## 1.3 Initial Project Plan

The team will follow an **Agile methodology** because the project still needs flexibility after the design stage. Agile is suitable here because it allows the team to lock the MVP boundary, review progress weekly, refine screen and flow details quickly, and introduce implementation changes without breaking the entire schedule.

**Key reasons for choosing Agile methodology:**

- **Flexibility:** The team expects some weekly requirement adjustments after design reviews.
- **Incremental Delivery:** Database, backend, and frontend work can be introduced in parallel after the design and ERD foundation is complete.
- **Risk Reduction:** High-risk modules such as checkout, kitchen queue logic, and delivery integration can be reviewed early.
- **User-Centered Design:** The team can validate screen flow and experience progressively instead of treating UI as a late-stage artifact.

### 1.3.1 Team Roles

- **Mai Van Nhat Minh – Technical Lead:** Owns architecture decisions, module boundaries, and integration review.
- **Ta Quoc Hung – Backend Lead:** Owns order flow, kitchen order queue logic, delivery integration, and core business rules.
- **Nguyen Xuan Chi Thanh – Product/UX Support:** Shares UI/UX work with the frontend lead, checks use-case coverage, and supports acceptance-flow consistency.
- **Ngo Manh Hieu – Database and Reporting Lead:** Owns ERD, schema design, migration direction, and reporting queries.
- **Tran Hoang – Frontend and UI Lead:** Owns the UI/UX direction, component inventory, responsive design, and frontend implementation strategy.

### 1.3.2 Roadmap and Staffing Plan

**Week 1 (23/03/2026 – 29/03/2026):** The team focuses on full UI/UX coverage for all MVP screens and finalizes ERD v1.

- **Minh:** Review screen flow and lock high-level module boundaries.
- **Hung:** Define order states, kitchen queue logic, and delivery handoff rules.
- **Thanh:** Co-design customer-facing UI/UX with Hoang and review use-case coverage.
- **Hieu:** Produce ERD v1 and the first data dictionary.
- **Hoang:** Lead the complete UI/UX design set and prototype preparation.

**Week 2 (30/03/2026 – 05/04/2026):** The team refines the design outputs and prepares technical execution artifacts.

- **Minh:** Finalize architecture, API conventions, and auth-role direction.
- **Hung:** Write flow specifications for checkout, kitchen queue, and delivery synchronization.
- **Thanh:** Refine screen naming, flow wording, and UX consistency with Hoang.
- **Hieu:** Upgrade ERD to v2 and map screens to database fields.
- **Hoang:** Finalize mockups, component inventory, and frontend design handoff.

**Weeks 3–4:** Build the project foundation, including auth, role setup, base schema, seed data, frontend shell, and menu/catalog groundwork.

**Weeks 5–6:** Implement the core ordering experience, including cart, COD checkout, guest tracking, customer account flow, loyalty points, and kitchen order queue behavior.

**Weeks 7–8:** Implement admin management, combo/category handling, delivery integration, basic reporting, and responsive refinements.

**Weeks 9–10:** Focus on integration testing, bug fixing, UAT, demo preparation, and release stabilization.

#### 1.3.3 CASE Tools

- **UML Diagram Creation:** Draw.io or Figma for use case diagrams, screen flows, and supporting visual models.
- **Database Design:** MySQL Workbench for ERD modeling and schema visualization.
- **Project Management:** Jira, Trello, or Notion for sprint tracking and weekly task management.
- **API Verification:** Postman or Bruno for API inspection and integration checks.

# 2 Analysis

## 2.1 General Use Case Overview

This section presents the use case analysis for the PizzaHút Online Ordering System. The scope is limited to a web-based MVP that can be delivered within 3.5 months while still satisfying the mandatory business requirements of the project topic. The refined MVP supports menu browsing, pizza customization, combo promotions, cash on delivery, guest order-code tracking, customer account features, loyalty points, kitchen order queue handling, third-party delivery integration, and basic reporting for the store owner.

### 2.1.1 General Use Case Diagram

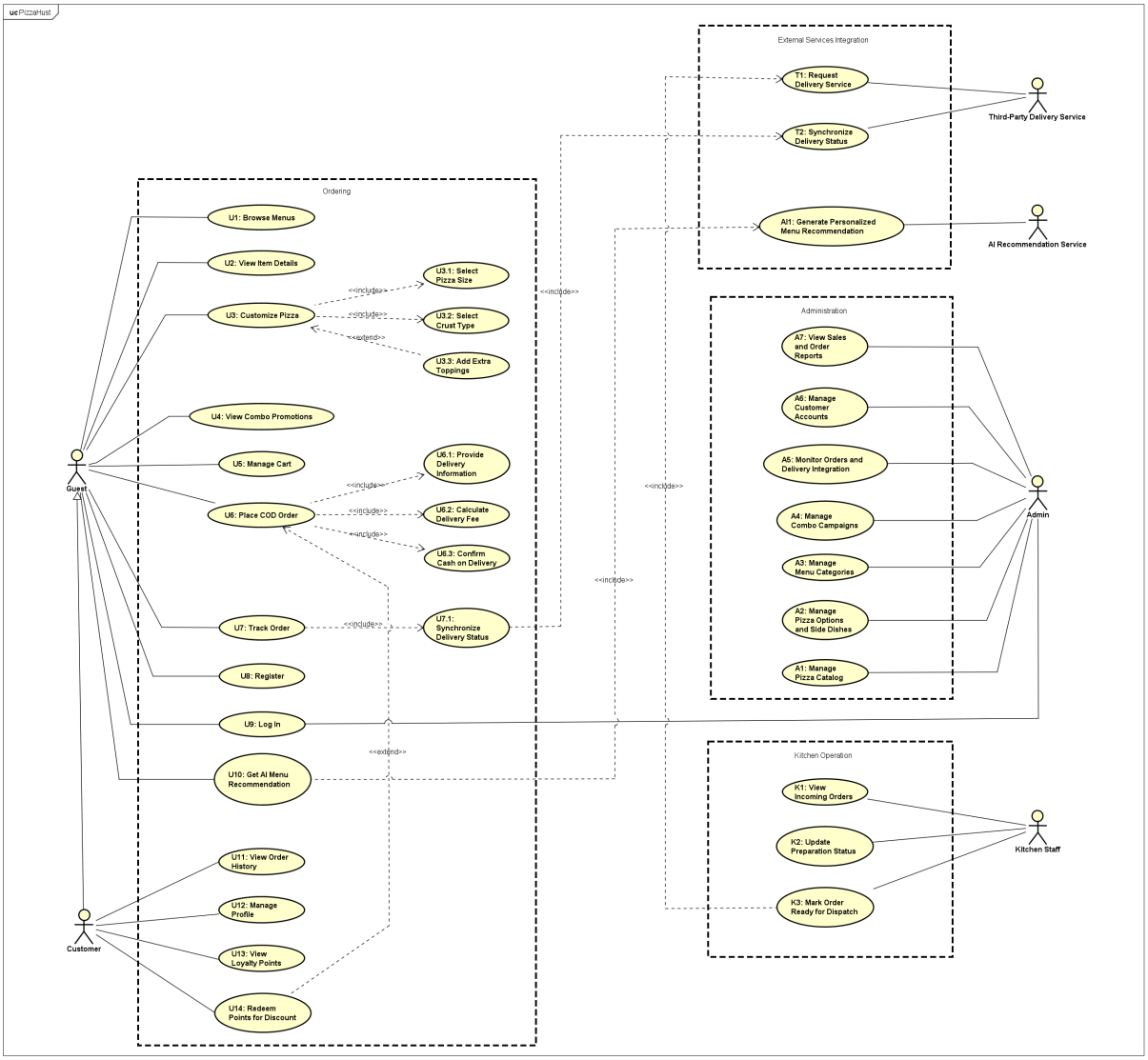


Figure 1: PizzaHút General Use Case Diagram

The figure above provides the high-level use case overview for PizzaHút. The detailed descriptions below follow the final MVP boundary used in this report. In particular, AI recommendation is treated as a future enhancement rather than an MVP use case, and

the kitchen workflow is refined into a **Kitchen Order Queue** that helps staff process the most urgent orders first.

The final MVP includes four primary internal actors and one external supporting actor:

**Guest:** A visitor who can browse menus, customize pizzas, place a cash-on-delivery order, and track an order by using a generated order code.

**Customer:** A registered user who inherits all guest capabilities and can additionally log in, manage profile information, view order history, earn loyalty points, and redeem points during checkout.

**Admin / Store Owner:** The business operator who manages pizzas, toppings, side dishes, categories, combo campaigns, customer records, order supervision, and sales reports.

**Kitchen Staff:** Internal staff members who work from a prioritized kitchen queue, process the next recommended order, update preparation progress, and mark completed orders as ready for dispatch.

**Third-Party Delivery Service:** An external delivery platform integrated through API to receive delivery requests and return delivery status updates.

### 2.1.2 Use Case Relationships

The main relationships among use cases are summarized below:

- **Customer** is modeled as a specialized actor that inherits all use cases available to **Guest**.
- **Customize Pizza** includes selecting pizza size and crust type, and it optionally extends to adding extra toppings.
- **Place COD Order** includes providing delivery information, calculating the fixed delivery fee, and confirming cash on delivery.
- **Redeem Points for Discount** extends **Place COD Order** and is available only to authenticated customers with sufficient loyalty points.
- **Track Order** includes **Synchronize Delivery Status** when the order has already been sent to the third-party delivery service.
- **Mark Order Ready for Dispatch** includes **Request Delivery Service** when the order is packed and ready for handoff.
- **Process Prioritized Order** depends on the prioritized kitchen queue that ranks waiting orders by urgency, promised time, and readiness constraints.

## 2.2 Use Case Details

### 2.2.1 Guest and Customer Use Case Details

**U1: Browse Menus**

<table>
<tr><td>Use Case Code</td><td>U1</td></tr>
<tr><td>Use Case Name</td><td>Browse Menus</td></tr>
<tr><td>Actor</td><td>Guest, Customer</td></tr>
<tr><td>Usage</td><td>Browse available items by menu category such as appetizer, main course, dessert, vegetarian, and kids menu</td></tr>
<tr><td>Activation Event</td><td>The user opens the website or navigates to the menu page</td></tr>
<tr><td>Preconditions</td><td>Active menu data must exist in the system</td></tr>
<tr><td>Main Event Flow</td><td><table>
<tr><th>Index</th><th>Done by</th><th>Action</th></tr>
<tr><td>1</td><td>User</td><td>Open the home page or menu page</td></tr>
<tr><td>2</td><td>System</td><td>Load available menu categories and featured items</td></tr>
<tr><td>3</td><td>User</td><td>Select a category or apply simple filters</td></tr>
<tr><td>4</td><td>System</td><td>Display matching products with prices and short descriptions</td></tr>
<tr><td>5</td><td>User</td><td>Continue browsing or open item details</td></tr>
</table></td></tr>
<tr><td>Postconditions</td><td>The user can view the products currently offered by the store</td></tr>
</table>

**U2: View Item Details**

<table>
<tr><td>Use Case Code</td><td>U2</td></tr>
<tr><td>Use Case Name</td><td>View Item Details</td></tr>
<tr><td>Actor</td><td>Guest, Customer</td></tr>
<tr><td>Usage</td><td>View detailed information about a pizza, side dish, or combo</td></tr>
<tr><td>Activation Event</td><td>The user selects an item from the menu</td></tr>
<tr><td>Preconditions</td><td>The selected item must be available in the catalog</td></tr>
<tr><td>Main Event Flow</td><td><table>
<tr><th>Index</th><th>Done by</th><th>Action</th></tr>
<tr><td>1</td><td>User</td><td>Select an item from the menu list</td></tr>
<tr><td>2</td><td>System</td><td>Load detailed product information</td></tr>
<tr><td>3</td><td>System</td><td>Display description, image, ingredients, and availability</td></tr>
<tr><td>4</td><td>System</td><td>Show pizza sizes, crust options, and toppings if the item is a pizza</td></tr>
<tr><td>5</td><td>User</td><td>Review the details before adding the item to the cart</td></tr>
</table></td></tr>
<tr><td>Postconditions</td><td>The user understands the selected item's details and purchase options</td></tr>
</table>

**U3: Customize Pizza**

<table>
<tr><td>Use Case Code</td><td>U3</td></tr>
<tr><td>Use Case Name</td><td>Customize Pizza</td></tr>
<tr><td>Actor</td><td>Guest, Customer</td></tr>
<tr><td>Usage</td><td>Customize a pizza by selecting size, crust type, and optional toppings</td></tr>
<tr><td>Activation Event</td><td>The user chooses to order a pizza</td></tr>
<tr><td>Preconditions</td><td>The selected pizza must be active and customizable</td></tr>
<tr><td>Main Event Flow</td><td><table>
<tr><th>Index</th><th>Done by</th><th>Action</th></tr>
<tr><td>1</td><td>User</td><td>Choose a pizza from the menu</td></tr>
<tr><td>2</td><td>System</td><td>Display available sizes and base prices</td></tr>
<tr><td>3</td><td>User</td><td>Select size S, M, or L</td></tr>
<tr><td>4</td><td>User</td><td>Select crust type</td></tr>
<tr><td>5</td><td>User</td><td>Optionally choose extra toppings</td></tr>
<tr><td>6</td><td>System</td><td>Recalculate the item price based on selected options</td></tr>
<tr><td>7</td><td>User</td><td>Confirm the customized pizza configuration</td></tr>
</table></td></tr>
<tr><td>Postconditions</td><td>A valid customized pizza is ready to be added to the cart</td></tr>
</table>

### U4: View Combo Promotions

<table>
<tr><td>Use Case Code</td><td>U4</td></tr>
<tr><td>Use Case Name</td><td>View Combo Promotions</td></tr>
<tr><td>Actor</td><td>Guest, Customer</td></tr>
<tr><td>Usage</td><td>View active combo bundles and promotional offers scheduled for different time windows</td></tr>
<tr><td>Activation Event</td><td>The user opens the combo or promotions page</td></tr>
<tr><td>Preconditions</td><td>At least one combo campaign must be active</td></tr>
<tr><td>Main Event Flow</td><td><table>
<tr><th>Index</th><th>Done by</th><th>Action</th></tr>
<tr><td>1</td><td>User</td><td>Open the combo or promotions section</td></tr>
<tr><td>2</td><td>System</td><td>Retrieve active combo campaigns based on the current time</td></tr>
<tr><td>3</td><td>System</td><td>Display combo contents, target group size, price, and validity period</td></tr>
<tr><td>4</td><td>User</td><td>Review available promotional options</td></tr>
<tr><td>5</td><td>User</td><td>Add a combo to the cart or continue browsing</td></tr>
</table></td></tr>
<tr><td>Postconditions</td><td>The user can evaluate available combo campaigns before ordering</td></tr>
</table>

### U5: Manage Cart

<table>
<tr><td>Use Case Code</td><td>U5</td></tr>
<tr><td>Use Case Name</td><td>Manage Cart</td></tr>
<tr><td>Actor</td><td>Guest, Customer</td></tr>
<tr><td>Usage</td><td>Add, update, or remove pizzas, side dishes, and combo items in the shopping cart</td></tr>
<tr><td>Activation Event</td><td>The user adds the first item to the cart or opens the cart page</td></tr>
<tr><td>Preconditions</td><td>At least one purchasable item must exist</td></tr>
<tr><td>Main Event Flow</td><td><table>
<tr><th>Index</th><th>Done by</th><th>Action</th></tr>
<tr><td>1</td><td>User</td><td>Add a customized pizza, side dish, or combo to the cart</td></tr>
<tr><td>2</td><td>System</td><td>Create or update the cart session</td></tr>
<tr><td>3</td><td>User</td><td>Change quantity or remove an item</td></tr>
<tr><td>4</td><td>System</td><td>Recalculate subtotal and current order total</td></tr>
<tr><td>5</td><td>User</td><td>Review the cart before checkout</td></tr>
</table></td></tr>
<tr><td>Postconditions</td><td>The cart contains the current set of items the user intends to purchase</td></tr>
</table>

**U6: Place COD Order**

<table>
<tr><td>Use Case Code</td><td>U6</td></tr>
<tr><td>Use Case Name</td><td>Place COD Order</td></tr>
<tr><td>Actor</td><td>Guest, Customer</td></tr>
<tr><td>Usage</td><td>Submit an online order and choose cash on delivery as the payment method</td></tr>
<tr><td>Activation Event</td><td>The user confirms checkout from the cart</td></tr>
<tr><td>Preconditions</td><td>The cart must contain at least one valid item</td></tr>
<tr><td>Main Event Flow</td><td><table>
<tr><th>Index</th><th>Done by</th><th>Action</th></tr>
<tr><td>1</td><td>User</td><td>Proceed to checkout from the cart</td></tr>
<tr><td>2</td><td>System</td><td>Display recipient and delivery information form</td></tr>
<tr><td>3</td><td>User</td><td>Enter recipient name, phone number, address, and delivery note</td></tr>
<tr><td>4</td><td>System</td><td>Apply the fixed inner Hanoi delivery fee of 22,000 VND</td></tr>
<tr><td>5</td><td>Customer</td><td>Optionally redeem loyalty points if authenticated</td></tr>
<tr><td>6</td><td>User</td><td>Confirm cash on delivery</td></tr>
<tr><td>7</td><td>System</td><td>Create the order and store all order lines</td></tr>
<tr><td>8</td><td>System</td><td>Generate an order code and place the order in the kitchen queue</td></tr>
</table></td></tr>
<tr><td>Postconditions</td><td>A new order is created with an initial processing status</td></tr>
</table>

**U7: Track Order**

<table>
<tr><td>Use Case Code</td><td>U7</td></tr>
<tr><td>Use Case Name</td><td>Track Order</td></tr>
<tr><td>Actor</td><td>Guest, Customer</td></tr>
<tr><td>Usage</td><td>Track the current order status after checkout</td></tr>
<tr><td>Activation Event</td><td>The user opens the tracking page or order detail page</td></tr>
<tr><td>Preconditions</td><td>A valid order code or authenticated customer session must exist</td></tr>
<tr><td>Main Event Flow</td><td>
<table>
<tr><th>Index</th><th>Done by</th><th>Action</th></tr>
<tr><td>1</td><td>User</td><td>Enter the order code or open an order from account history</td></tr>
<tr><td>2</td><td>System</td><td>Retrieve the current order record</td></tr>
<tr><td>3</td><td>System</td><td>Display status timeline such as Received, Preparing, Ready for Dispatch, Delivering, Delivered, or Delivery Failed</td></tr>
<tr><td>4</td><td>System</td><td>Synchronize delivery status if an external delivery reference exists</td></tr>
<tr><td>5</td><td>User</td><td>Review the latest order progress</td></tr>
</table>
</td></tr>
<tr><td>Postconditions</td><td>The user knows the current status of the order</td></tr>
</table>

**U8: Register**

<table>
<tr><td>Use Case Code</td><td>U8</td></tr>
<tr><td>Use Case Name</td><td>Register</td></tr>
<tr><td>Actor</td><td>Guest</td></tr>
<tr><td>Usage</td><td>Create a customer account to use history and loyalty features</td></tr>
<tr><td>Activation Event</td><td>The guest chooses the register option</td></tr>
<tr><td>Preconditions</td><td>The guest must not already be authenticated</td></tr>
<tr><td>Main Event Flow</td><td>
<table>
<tr><th>Index</th><th>Done by</th><th>Action</th></tr>
<tr><td>1</td><td>Guest</td><td>Open the registration page</td></tr>
<tr><td>2</td><td>System</td><td>Display the registration form</td></tr>
<tr><td>3</td><td>Guest</td><td>Enter required account information</td></tr>
<tr><td>4</td><td>System</td><td>Validate the submitted data and check uniqueness</td></tr>
<tr><td>5</td><td>System</td><td>Create the customer account and loyalty profile</td></tr>
<tr><td>6</td><td>System</td><td>Notify successful registration</td></tr>
</table>
</td></tr>
<tr><td>Postconditions</td><td>A new customer account is available for future logins and point accumulation</td></tr>
</table>

**U9: Log In**

| | |
|---|---|
| **Use Case Code** | U9 |
| **Use Case Name** | Log In |
| **Actor** | Customer |
| **Usage** | Authenticate a registered customer and open an account session |
| **Activation Event** | The customer chooses the log in option |
| **Preconditions** | The customer account must already exist |
| **Main Event Flow** | |

| Index | Done by | Action |
|---|---|---|
| 1 | Customer | Open the log in page |
| 2 | System | Display the authentication form |
| 3 | Customer | Submit email or phone number and password |
| 4 | System | Validate the credentials |
| 5 | System | Create the authenticated session and redirect to the account area |

| | |
|---|---|
| **Postconditions** | The customer is logged in and can access account-only features |

**U10: View Order History**

| | |
|---|---|
| **Use Case Code** | U10 |
| **Use Case Name** | View Order History |
| **Actor** | Customer |
| **Usage** | Review previous orders placed with the authenticated account |
| **Activation Event** | The customer opens the order history page |
| **Preconditions** | The customer must be logged in |
| **Main Event Flow** | |

| Index | Done by | Action |
|---|---|---|
| 1 | Customer | Open the order history page |
| 2 | System | Retrieve past orders associated with the account |
| 3 | System | Display order codes, dates, totals, and statuses |
| 4 | Customer | Select an order to view details |

| | |
|---|---|
| **Postconditions** | The customer can review previous purchase activities |

**U11: Manage Profile**

<table>
<tr><td>Use Case Code</td><td>U11</td></tr>
<tr><td>Use Case Name</td><td>Manage Profile</td></tr>
<tr><td>Actor</td><td>Customer</td></tr>
<tr><td>Usage</td><td>Update personal information for future orders</td></tr>
<tr><td>Activation Event</td><td>The customer opens the profile page</td></tr>
<tr><td>Preconditions</td><td>The customer must be logged in</td></tr>
<tr><td>Main Event Flow</td><td><table>
<tr><th>Index</th><th>Done by</th><th>Action</th></tr>
<tr><td>1</td><td>Customer</td><td>Open the profile management page</td></tr>
<tr><td>2</td><td>System</td><td>Display current account information</td></tr>
<tr><td>3</td><td>Customer</td><td>Update name, phone number, password, or default address</td></tr>
<tr><td>4</td><td>System</td><td>Validate and save the changes</td></tr>
<tr><td>5</td><td>System</td><td>Show a success confirmation</td></tr>
</table></td></tr>
<tr><td>Postconditions</td><td>The customer profile is updated for future account usage</td></tr>
</table>

**U12: View Loyalty Points**

<table>
<tr><td>Use Case Code</td><td>U12</td></tr>
<tr><td>Use Case Name</td><td>View Loyalty Points</td></tr>
<tr><td>Actor</td><td>Customer</td></tr>
<tr><td>Usage</td><td>View the current loyalty point balance and recent point transactions</td></tr>
<tr><td>Activation Event</td><td>The customer opens the loyalty page</td></tr>
<tr><td>Preconditions</td><td>The customer must be logged in</td></tr>
<tr><td>Main Event Flow</td><td><table>
<tr><th>Index</th><th>Done by</th><th>Action</th></tr>
<tr><td>1</td><td>Customer</td><td>Open the loyalty page</td></tr>
<tr><td>2</td><td>System</td><td>Retrieve the current point balance and recent point movements</td></tr>
<tr><td>3</td><td>System</td><td>Display the balance and current redemption rules</td></tr>
<tr><td>4</td><td>Customer</td><td>Review loyalty information</td></tr>
</table></td></tr>
<tr><td>Postconditions</td><td>The customer understands the current loyalty point status</td></tr>
</table>

**U13: Redeem Points for Discount**

<table>
<tr><td>Use Case Code</td><td>U13</td></tr>
<tr><td>Use Case Name</td><td>Redeem Points for Discount</td></tr>
<tr><td>Actor</td><td>Customer</td></tr>
<tr><td>Usage</td><td>Use loyalty points to reduce the final order value during checkout</td></tr>
<tr><td>Activation Event</td><td>The logged-in customer reaches the checkout step</td></tr>
<tr><td>Preconditions</td><td>The customer must be logged in and have sufficient points</td></tr>
<tr><td>Main Event Flow</td><td>
<table>
<tr><th>Index</th><th>Done by</th><th>Action</th></tr>
<tr><td>1</td><td>Customer</td><td>Choose to apply loyalty points during checkout</td></tr>
<tr><td>2</td><td>System</td><td>Check the current point balance and applicable discount rule</td></tr>
<tr><td>3</td><td>Customer</td><td>Select the amount of points to redeem</td></tr>
<tr><td>4</td><td>System</td><td>Convert points into a valid discount amount</td></tr>
<tr><td>5</td><td>System</td><td>Update the order total before final confirmation</td></tr>
<tr><td>6</td><td>Customer</td><td>Continue the COD checkout process</td></tr>
</table>
</td></tr>
<tr><td>Postconditions</td><td>The order uses the selected loyalty discount and reserves the redeemed points</td></tr>
</table>

### 2.2.2 Admin / Store Owner Use Case Details

**A1: Manage Pizza Catalog**

<table>
<tr><td>Use Case Code</td><td>A1</td></tr>
<tr><td>Use Case Name</td><td>Manage Pizza Catalog</td></tr>
<tr><td>Actor</td><td>Admin / Store Owner</td></tr>
<tr><td>Usage</td><td>Create, update, activate, or deactivate pizza products in the catalog</td></tr>
<tr><td>Activation Event</td><td>The admin opens the pizza management page</td></tr>
<tr><td>Preconditions</td><td>The admin must be authenticated with management privileges</td></tr>
<tr><td>Main Event Flow</td><td>
<table>
<tr><th>Index</th><th>Done by</th><th>Action</th></tr>
<tr><td>1</td><td>Admin</td><td>Open the pizza catalog management page</td></tr>
<tr><td>2</td><td>System</td><td>Display existing pizza products</td></tr>
<tr><td>3</td><td>Admin</td><td>Create a new pizza or edit an existing one</td></tr>
<tr><td>4</td><td>Admin</td><td>Update name, description, image, and active status</td></tr>
<tr><td>5</td><td>System</td><td>Save the catalog changes</td></tr>
</table>
</td></tr>
<tr><td>Postconditions</td><td>The pizza catalog reflects the latest business offerings</td></tr>
</table>

**A2: Manage Pizza Options and Side Dishes**

<table>
<tr><td>Use Case Code</td><td>A2</td></tr>
<tr><td>Use Case Name</td><td>Manage Pizza Options and Side Dishes</td></tr>
<tr><td>Actor</td><td>Admin / Store Owner</td></tr>
<tr><td>Usage</td><td>Manage pizza sizes, size-based prices, crust types, toppings, topping prices, and side dish products</td></tr>
<tr><td>Activation Event</td><td>The admin opens the product options management page</td></tr>
<tr><td>Preconditions</td><td>The admin must be authenticated with management privileges</td></tr>
<tr><td>Main Event Flow</td><td><table>
<tr><th>Index</th><th>Done by</th><th>Action</th></tr>
<tr><td>1</td><td>Admin</td><td>Open the options and side dish management page</td></tr>
<tr><td>2</td><td>System</td><td>Display sizes, crust types, toppings, and side dish items</td></tr>
<tr><td>3</td><td>Admin</td><td>Update size-specific prices and topping prices</td></tr>
<tr><td>4</td><td>Admin</td><td>Add, edit, activate, or deactivate side dish products</td></tr>
<tr><td>5</td><td>System</td><td>Save the updated configuration</td></tr>
</table></td></tr>
<tr><td>Postconditions</td><td>Item options and side dish data are ready for customer ordering</td></tr>
</table>

### A3: Manage Menu Categories

<table>
<tr><td>Use Case Code</td><td>A3</td></tr>
<tr><td>Use Case Name</td><td>Manage Menu Categories</td></tr>
<tr><td>Actor</td><td>Admin / Store Owner</td></tr>
<tr><td>Usage</td><td>Maintain the menu structure such as appetizer, main course, dessert, vegetarian, and kids menu</td></tr>
<tr><td>Activation Event</td><td>The admin opens category management</td></tr>
<tr><td>Preconditions</td><td>The admin must be authenticated with management privileges</td></tr>
<tr><td>Main Event Flow</td><td><table>
<tr><th>Index</th><th>Done by</th><th>Action</th></tr>
<tr><td>1</td><td>Admin</td><td>Open the menu category management page</td></tr>
<tr><td>2</td><td>System</td><td>Display existing categories and item assignments</td></tr>
<tr><td>3</td><td>Admin</td><td>Create, edit, reorder, or deactivate categories</td></tr>
<tr><td>4</td><td>Admin</td><td>Assign products to one or more categories</td></tr>
<tr><td>5</td><td>System</td><td>Save the updated menu structure</td></tr>
</table></td></tr>
<tr><td>Postconditions</td><td>The website shows menu categories in the expected business structure</td></tr>
</table>

### A4: Manage Combo Campaigns

<table>
<tr><td>Use Case Code</td><td>A4</td></tr>
<tr><td>Use Case Name</td><td>Manage Combo Campaigns</td></tr>
<tr><td>Actor</td><td>Admin / Store Owner</td></tr>
<tr><td>Usage</td><td>Create and schedule combo promotions for lunch, dinner, or group-based offers</td></tr>
<tr><td>Activation Event</td><td>The admin opens combo management</td></tr>
<tr><td>Preconditions</td><td>The admin must be authenticated with management privileges</td></tr>
<tr><td>Main Event Flow</td><td><table>
<tr><th>Index</th><th>Done by</th><th>Action</th></tr>
<tr><td>1</td><td>Admin</td><td>Open the combo campaign page</td></tr>
<tr><td>2</td><td>System</td><td>Display active, scheduled, and expired combo records</td></tr>
<tr><td>3</td><td>Admin</td><td>Create or edit combo contents, price, validity period, and target customer group</td></tr>
<tr><td>4</td><td>System</td><td>Validate the campaign data and price consistency</td></tr>
<tr><td>5</td><td>System</td><td>Save and publish the campaign</td></tr>
</table></td></tr>
<tr><td>Postconditions</td><td>Active combo promotions are available for customer browsing and ordering</td></tr>
</table>

### A5: Monitor Orders and Delivery Exceptions

<table>
<tr><td>Use Case Code</td><td>A5</td></tr>
<tr><td>Use Case Name</td><td>Monitor Orders and Delivery Exceptions</td></tr>
<tr><td>Actor</td><td>Admin / Store Owner</td></tr>
<tr><td>Usage</td><td>Monitor the order pipeline and handle delivery or order exceptions that need manual supervision</td></tr>
<tr><td>Activation Event</td><td>The admin opens the order dashboard</td></tr>
<tr><td>Preconditions</td><td>The admin must be authenticated with management privileges</td></tr>
<tr><td>Main Event Flow</td><td><table>
<tr><th>Index</th><th>Done by</th><th>Action</th></tr>
<tr><td>1</td><td>Admin</td><td>Open the order monitoring dashboard</td></tr>
<tr><td>2</td><td>System</td><td>Display orders by status, delivery state, and exception flag</td></tr>
<tr><td>3</td><td>Admin</td><td>Review an order that needs manual support or delivery follow-up</td></tr>
<tr><td>4</td><td>System</td><td>Show kitchen progress, delivery reference, and latest delivery state</td></tr>
<tr><td>5</td><td>Admin</td><td>Retry or supervise the delivery request when an exception occurs</td></tr>
<tr><td>6</td><td>System</td><td>Record the intervention and refresh tracking data</td></tr>
</table></td></tr>
<tr><td>Postconditions</td><td>Order exceptions can be resolved without blocking the main workflow</td></tr>
</table>

### A6: Manage Customer Accounts

<table>
<tr><td>Use Case Code</td><td>A6</td></tr>
<tr><td>Use Case Name</td><td>Manage Customer Accounts</td></tr>
<tr><td>Actor</td><td>Admin / Store Owner</td></tr>
<tr><td>Usage</td><td>View customer records and handle account-related support tasks</td></tr>
<tr><td>Activation Event</td><td>The admin opens the customer management page</td></tr>
<tr><td>Preconditions</td><td>The admin must be authenticated with management privileges</td></tr>
<tr><td>Main Event Flow</td><td><table>
<tr><th>Index</th><th>Done by</th><th>Action</th></tr>
<tr><td>1</td><td>Admin</td><td>Open the customer management page</td></tr>
<tr><td>2</td><td>System</td><td>Display customer records, loyalty balances, and account status</td></tr>
<tr><td>3</td><td>Admin</td><td>Search or inspect a customer account</td></tr>
<tr><td>4</td><td>Admin</td><td>Perform support actions such as updating status or assisting with loyalty issues</td></tr>
<tr><td>5</td><td>System</td><td>Save the administrative changes</td></tr>
</table></td></tr>
<tr><td>Postconditions</td><td>Customer account records remain manageable from the administration side</td></tr>
</table>

**A7: View Sales and Order Reports**

<table>
<tr><td>Use Case Code</td><td>A7</td></tr>
<tr><td>Use Case Name</td><td>View Sales and Order Reports</td></tr>
<tr><td>Actor</td><td>Admin / Store Owner</td></tr>
<tr><td>Usage</td><td>Review basic business reports for sales, order volume, top items, and delivery performance</td></tr>
<tr><td>Activation Event</td><td>The admin opens the reporting dashboard</td></tr>
<tr><td>Preconditions</td><td>Order data must exist in the system</td></tr>
<tr><td>Main Event Flow</td><td><table>
<tr><th>Index</th><th>Done by</th><th>Action</th></tr>
<tr><td>1</td><td>Admin</td><td>Open the reporting dashboard</td></tr>
<tr><td>2</td><td>System</td><td>Request a date range or report type</td></tr>
<tr><td>3</td><td>Admin</td><td>Select the reporting criteria</td></tr>
<tr><td>4</td><td>System</td><td>Aggregate revenue, order counts, best-selling items, and delivery metrics</td></tr>
<tr><td>5</td><td>System</td><td>Display the report in charts and summary tables</td></tr>
</table></td></tr>
<tr><td>Postconditions</td><td>The admin can assess business performance using basic operational reports</td></tr>
</table>

### 2.2.3 Kitchen Staff Use Case Details

**K1: View Kitchen Order Queue**

<table>
<tr><td>Use Case Code</td><td>K1</td></tr>
<tr><td>Use Case Name</td><td>View Kitchen Order Queue</td></tr>
<tr><td>Actor</td><td>Kitchen Staff</td></tr>
<tr><td>Usage</td><td>View the prioritized queue of waiting and active orders in the kitchen</td></tr>
<tr><td>Activation Event</td><td>Kitchen staff opens the kitchen dashboard</td></tr>
<tr><td>Preconditions</td><td>At least one order must exist in the kitchen workflow</td></tr>
<tr><td>Main Event Flow</td><td><table>
<tr><th>Index</th><th>Done by</th><th>Action</th></tr>
<tr><td>1</td><td>Kitchen Staff</td><td>Open the kitchen dashboard</td></tr>
<tr><td>2</td><td>System</td><td>Calculate queue priority from order time, promised time, and current status</td></tr>
<tr><td>3</td><td>System</td><td>Display waiting and active kitchen orders in recommended processing order</td></tr>
<tr><td>4</td><td>Kitchen Staff</td><td>Review the next suggested order and inspect details</td></tr>
</table></td></tr>
<tr><td>Postconditions</td><td>Kitchen staff knows which orders should be handled first</td></tr>
</table>

### K2: Process Prioritized Order

<table>
<tr><td>Use Case Code</td><td>K2</td></tr>
<tr><td>Use Case Name</td><td>Process Prioritized Order</td></tr>
<tr><td>Actor</td><td>Kitchen Staff</td></tr>
<tr><td>Usage</td><td>Accept and start working on the next recommended order from the queue</td></tr>
<tr><td>Activation Event</td><td>Kitchen staff chooses the next suggested order</td></tr>
<tr><td>Preconditions</td><td>A prioritized order must be available in the kitchen queue</td></tr>
<tr><td>Main Event Flow</td><td><table>
<tr><th>Index</th><th>Done by</th><th>Action</th></tr>
<tr><td>1</td><td>Kitchen Staff</td><td>Select the next prioritized order</td></tr>
<tr><td>2</td><td>System</td><td>Display item list, pizza customizations, and customer notes</td></tr>
<tr><td>3</td><td>Kitchen Staff</td><td>Confirm that the order has started preparation</td></tr>
<tr><td>4</td><td>System</td><td>Mark the order as actively processed in the kitchen</td></tr>
</table></td></tr>
<tr><td>Postconditions</td><td>The selected order becomes the current kitchen work item</td></tr>
</table>

### K3: Update Preparation Status

<table>
<tr><td>Use Case Code</td><td>K3</td></tr>
<tr><td>Use Case Name</td><td>Update Preparation Status</td></tr>
<tr><td>Actor</td><td>Kitchen Staff</td></tr>
<tr><td>Usage</td><td>Update the current preparation stage of an order</td></tr>
<tr><td>Activation Event</td><td>Kitchen staff starts or progresses work on an active order</td></tr>
<tr><td>Preconditions</td><td>The target order must exist in the kitchen workflow</td></tr>
<tr><td>Main Event Flow</td><td><table>
<tr><th>Index</th><th>Done by</th><th>Action</th></tr>
<tr><td>1</td><td>Kitchen Staff</td><td>Open an active kitchen order</td></tr>
<tr><td>2</td><td>Kitchen Staff</td><td>Set status such as Preparing or Baking</td></tr>
<tr><td>3</td><td>System</td><td>Record the new preparation status and timestamp</td></tr>
<tr><td>4</td><td>System</td><td>Update customer tracking and admin monitoring data</td></tr>
</table></td></tr>
<tr><td>Postconditions</td><td>The current preparation state of the order is visible to related actors</td></tr>
</table>

**K4: Mark Order Ready for Dispatch**

<table>
<tr><td>Use Case Code</td><td>K4</td></tr>
<tr><td>Use Case Name</td><td>Mark Order Ready for Dispatch</td></tr>
<tr><td>Actor</td><td>Kitchen Staff</td></tr>
<tr><td>Usage</td><td>Confirm that a prepared order is packed and ready to be handed over to the integrated delivery service</td></tr>
<tr><td>Activation Event</td><td>Kitchen staff finishes preparing an order</td></tr>
<tr><td>Preconditions</td><td>The target order must be completed by the kitchen</td></tr>
<tr><td>Main Event Flow</td><td><table>
<tr><th>Index</th><th>Done by</th><th>Action</th></tr>
<tr><td>1</td><td>Kitchen Staff</td><td>Open a completed kitchen order</td></tr>
<tr><td>2</td><td>Kitchen Staff</td><td>Confirm that the order is packed and ready</td></tr>
<tr><td>3</td><td>System</td><td>Change the order status to Ready for Dispatch</td></tr>
<tr><td>4</td><td>System</td><td>Trigger a delivery request to the third-party delivery service</td></tr>
<tr><td>5</td><td>System</td><td>Store the delivery reference and update the tracking timeline</td></tr>
</table></td></tr>
<tr><td>Postconditions</td><td>The order is handed off from kitchen workflow to the external delivery workflow</td></tr>
</table>

### 2.2.4 Third-Party Delivery Service Use Case Details

**T1: Request Delivery Service**

<table>
<tr><td>Use Case Code</td><td>T1</td></tr>
<tr><td>Use Case Name</td><td>Request Delivery Service</td></tr>
<tr><td>Actor</td><td>Third-Party Delivery Service</td></tr>
<tr><td>Usage</td><td>Receive a delivery booking request from the system and return a tracking reference</td></tr>
<tr><td>Activation Event</td><td>The system sends a delivery request after the kitchen marks an order as ready for dispatch</td></tr>
<tr><td>Preconditions</td><td>The order must be packed, have valid delivery information, and be ready for dispatch</td></tr>
<tr><td>Main Event Flow</td><td>
<table>
<tr><th>Index</th><th>Done by</th><th>Action</th></tr>
<tr><td>1</td><td>System</td><td>Send order pickup, drop-off, contact, and COD information to the delivery API</td></tr>
<tr><td>2</td><td>Third-Party Delivery Service</td><td>Validate the request and check service availability</td></tr>
<tr><td>3</td><td>Third-Party Delivery Service</td><td>Create a delivery job and generate a tracking reference</td></tr>
<tr><td>4</td><td>Third-Party Delivery Service</td><td>Return the booking result to the system</td></tr>
<tr><td>5</td><td>System</td><td>Save the delivery reference and update order tracking data</td></tr>
</table>
</td></tr>
<tr><td>Postconditions</td><td>The order is successfully submitted to the external delivery service for fulfillment</td></tr>
</table>

**T2: Synchronize Delivery Status**

<table>
<tr><td>Use Case Code</td><td>T2</td></tr>
<tr><td>Use Case Name</td><td>Synchronize Delivery Status</td></tr>
<tr><td>Actor</td><td>Third-Party Delivery Service</td></tr>
<tr><td>Usage</td><td>Send updated delivery progress from the external delivery platform back to the ordering system</td></tr>
<tr><td>Activation Event</td><td>The external delivery platform changes the delivery state of an order</td></tr>
<tr><td>Preconditions</td><td>A valid delivery reference must already exist in both systems</td></tr>
<tr><td>Main Event Flow</td><td>
<table>
<tr><th>Index</th><th>Done by</th><th>Action</th></tr>
<tr><td>1</td><td>Third-Party Delivery Service</td><td>Change the delivery state such as Accepted, Delivering, Delivered, or Failed</td></tr>
<tr><td>2</td><td>Third-Party Delivery Service</td><td>Send the latest delivery status to the system through API or webhook</td></tr>
<tr><td>3</td><td>System</td><td>Match the update with the internal order using the delivery reference</td></tr>
<tr><td>4</td><td>System</td><td>Update the order tracking timeline and administrative monitoring data</td></tr>
<tr><td>5</td><td>System</td><td>Expose the latest delivery progress to customers on the tracking page</td></tr>
</table>
</td></tr>
<tr><td>Postconditions</td><td>The local system stays consistent with the latest delivery information from the external platform</td></tr>
</table>